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EMBEDDED AI

Real-Time Fastener Detection on Raspberry Pi

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Contents

A typical slide

- Write here the introduction
- Use `itemize` or `enumerate` to organize your main points.

Example

Rather than freely writing text in the frame, the use of blocks (such as this) is recommended. Here you can find a short introduction to the basis of beamer:

1. https://www.overleaf.com/learn/latex/beamer

Workflow

- Set up Raspberry Pi 4/5:
  - Install OS
  - Connect to the Wi-Fi network
  - Install drivers for the camera - IDS SDK
- Record videos via the camera, connected to Raspberry Pi
- Annotate items via Roboflow framework
- Run training on GPU
- Test real-time detection on Raspberry Pi

Camera (UI-3590CP-C-HQ Rev.2)

Sensor	
Resolution	18.10 Mpix (4912×3684)
Gain (master/RGB)	16× / 4×
Model	
Pixel clock range	96–432 MHz
Max frame rate (freerun/trigger)	21 / 21 fps
Exposure time (min–max)	0.013–998 ms
Power consumption	1.5–3.3 W
Image memory	128 MB
Environment & I/O	
Interface	USB 3.0 micro-B, screwable

Roboflow

Roboflow is a platform for rapid image annotation, with APIs to upload images and export datasets in model-ready formats